

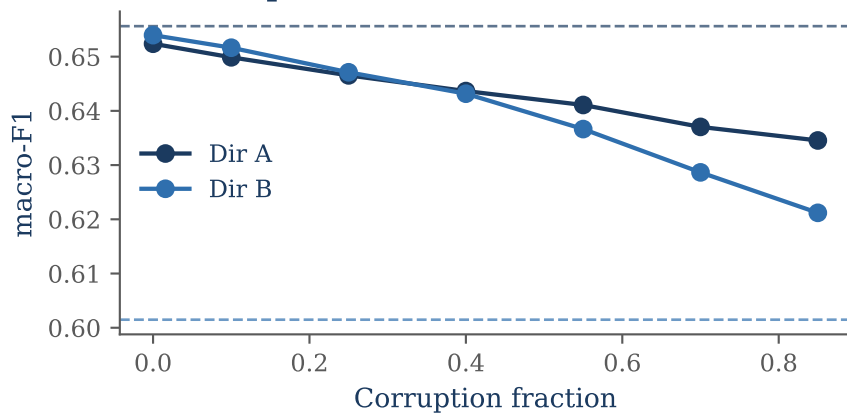
A. Training-time degradation

Corrupt training protein → retrain totalVI → evaluate
(scVI fixed reference)
Within-dataset HC development setting

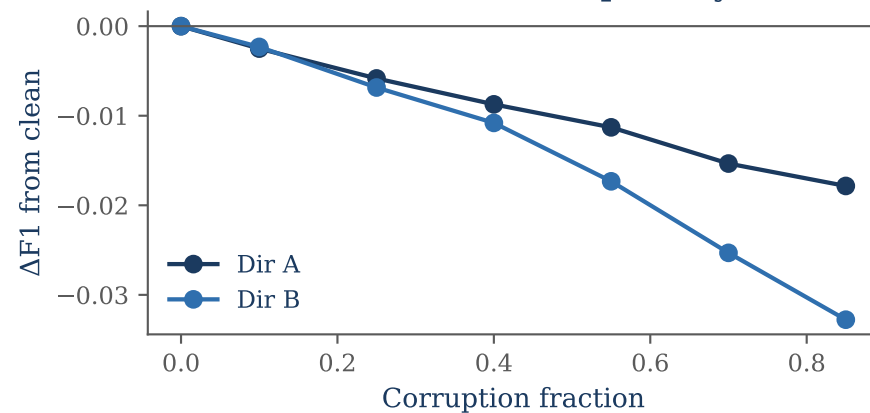
B. Post-adaptation target-only degradation

Clean source train → clean scArches adapt → freeze
→ corrupt TARGET protein only → re-infer
Primary endpoint: Δ from clean within seed

C-D. Post-adaptation totalVI F1 (dashed = clean scVI)



E. Δ F1 from clean (primary)



F. Most degradation-sensitive cell types (Direction B)

